Mathematics for Data Science and Machine Learning: Differential Calculus

Slopes and Rate of Change

EASY 1. Calculate the slope m for the segment connecting (x1, y1) = (1, 1) and (x2, y2) = (3, 6).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 2. Calculate the slope m for the segment connecting (x1, y1) = (1, 2) and (x2, y2) = (3, 7).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 3. Calculate the slope m for the segment connecting (x1, y1) = (1, 3) and (x2, y2) = (3, 8).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 4. Calculate the slope m for the segment connecting (x1, y1) = (1, 4) and (x2, y2) = (3, 9).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 5. Calculate the slope m for the segment connecting (x1, y1) = (1, 5) and (x2, y2) = (3, 10).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 6. Calculate the slope m for the segment connecting (x1, y1) = (1, 6) and (x2, y2) = (3, 11).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 7. Calculate the slope m for the segment connecting (x1, y1) = (1, 7) and (x2, y2) = (3, 12).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

EASY 8. Calculate the slope m for the segment connecting (x1, y1) = (1, 8) and (x2, y2) = (3, 13).

Rule/Reference: `m = (y2 - y1) / (x2 - x1)`

INTERM. 9. Find the secant slope for $f(x) = 9x^2 + 1$ on the interval [8, 9].

Rule/Reference: `m_sec = (f(x2) - f(x1)) / (x2 - x1)`

INTERM. 10. Find the secant slope for $f(x) = 10x^2 + 1$ on the interval [9, 10].

Rule/Reference: `m_sec = (f(x2) - f(x1)) / (x2 - x1)`

INTERM. 11. Find the secant slope for $f(x) = 11x^2 + 1$ on the interval [10, 11].

Rule/Reference: `m_sec = (f(x2) - f(x1)) / (x2 - x1)`

INTERM. 12. Find the secant slope for $f(x) = 12x^2 + 1$ on the interval [11, 12].

Rule/Reference: `m_sec = (f(x2) - f(x1)) / (x2 - x1)`

INTERM. 13. Find the secant slope for $f(x) = 13x^2 + 1$ on the interval [12, 13].

Rule/Reference: m_sec = (f(x2) - f(x1)) / (x2 - x1)

INTERM. 14. Find the secant slope for $f(x) = 14x^2 + 1$ on the interval [13, 14].

Rule/Reference: m_sec = (f(x2) - f(x1)) / (x2 - x1)

INTERM. 15. Find the secant slope for $f(x) = 15x^2 + 1$ on the interval [14, 15].

Rule/Reference: m_sec = (f(x2) - f(x1)) / (x2 - x1)

INTERM. 16. Find the secant slope for $f(x) = 16x^2 + 1$ on the interval [15, 16].

Rule/Reference: m_sec = (f(x2) - f(x1)) / (x2 - x1)

INTERM. 17. Find the secant slope for $f(x) = 17x^2 + 1$ on the interval [16, 17].

Rule/Reference: m_sec = (f(x2) - f(x1)) / (x2 - x1)

DIFFICULT 18. Find the instantaneous rate of change of $f(x) = 18x^3 - x$ at $x = 2$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 19. Find the instantaneous rate of change of $f(x) = 19x^3 - x$ at $x = 3$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 20. Find the instantaneous rate of change of $f(x) = 20x^3 - x$ at $x = 4$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 21. Find the instantaneous rate of change of $f(x) = 21x^3 - x$ at $x = 0$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 22. Find the instantaneous rate of change of $f(x) = 22x^3 - x$ at $x = 1$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 23. Find the instantaneous rate of change of $f(x) = 23x^3 - x$ at $x = 2$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 24. Find the instantaneous rate of change of $f(x) = 24x^3 - x$ at $x = 3$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

DIFFICULT 25. Find the instantaneous rate of change of $f(x) = 25x^3 - x$ at $x = 4$.

Rule/Reference: f'(a) = lim(h -> 0) [f(a+h) - f(a)] / h

Introduction to Derivatives

EASY 1. Interpret the rate of change for $f(x) = 1x + 0$.

Rule/Reference: f'(x) = m

EASY 2. Interpret the rate of change for $f(x) = 2x + 1$.

Rule/Reference: `f'(x) = m`

EASY 3. Interpret the rate of change for $f(x) = 3x + 2$.

Rule/Reference: `f'(x) = m`

EASY 4. Interpret the rate of change for $f(x) = 4x + 3$.

Rule/Reference: `f'(x) = m`

EASY 5. Interpret the rate of change for $f(x) = 5x + 4$.

Rule/Reference: `f'(x) = m`

EASY 6. Interpret the rate of change for $f(x) = 6x + 5$.

Rule/Reference: `f'(x) = m`

EASY 7. Interpret the rate of change for $f(x) = 7x + 6$.

Rule/Reference: `f'(x) = m`

EASY 8. Interpret the rate of change for $f(x) = 8x + 7$.

Rule/Reference: `f'(x) = m`

INTERM. 9. Interpret the rate of change for $f(x) = 9x + 8$.

Rule/Reference: `f'(x) = m`

INTERM. 10. Interpret the rate of change for $f(x) = 10x + 9$.

Rule/Reference: `f'(x) = m`

INTERM. 11. Interpret the rate of change for $f(x) = 11x + 10$.

Rule/Reference: `f'(x) = m`

INTERM. 12. Interpret the rate of change for $f(x) = 12x + 11$.

Rule/Reference: `f'(x) = m`

INTERM. 13. Interpret the rate of change for $f(x) = 13x + 12$.

Rule/Reference: `f'(x) = m`

INTERM. 14. Interpret the rate of change for $f(x) = 14x + 13$.

Rule/Reference: `f'(x) = m`

INTERM. 15. Interpret the rate of change for $f(x) = 15x + 14$.

Rule/Reference: `f'(x) = m`

INTERM. 16. Interpret the rate of change for $f(x) = 16x + 15$.

Rule/Reference: `f'(x) = m`

INTERM. 17. Interpret the rate of change for $f(x) = 17x + 16$.

Rule/Reference: `f'(x) = m`

DIFFICULT 18. Interpret the rate of change for f(x) = 18x + 17.

Rule/Reference: f'(x) = m

DIFFICULT 19. Interpret the rate of change for f(x) = 19x + 18.

Rule/Reference: f'(x) = m

DIFFICULT 20. Interpret the rate of change for f(x) = 20x + 19.

Rule/Reference: f'(x) = m

DIFFICULT 21. Interpret the rate of change for f(x) = 21x + 20.

Rule/Reference: f'(x) = m

DIFFICULT 22. Interpret the rate of change for f(x) = 22x + 21.

Rule/Reference: f'(x) = m

DIFFICULT 23. Interpret the rate of change for f(x) = 23x + 22.

Rule/Reference: f'(x) = m

DIFFICULT 24. Interpret the rate of change for f(x) = 24x + 23.

Rule/Reference: f'(x) = m

DIFFICULT 25. Interpret the rate of change for f(x) = 25x + 24.

Rule/Reference: f'(x) = m

Mathematical Notation with Limits

EASY 1. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 1x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 2. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 2x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 3. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 3x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 4. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 4x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 5. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 5x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 6. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = 6x^2.

Rule/Reference: f'(x) = 2({idx})x

EASY 7. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 7x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

EASY 8. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 8x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 9. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 9x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 10. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 10x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 11. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 11x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 12. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 12x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 13. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 13x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 14. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 14x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 15. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 15x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 16. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 16x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

INTERM. 17. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 17x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 18. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 18x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 19. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 19x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 20. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for $f(x) = 20x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 21. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = $21x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 22. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = $22x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 23. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = $23x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 24. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = $24x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

DIFFICULT 25. Evaluate lim(h -> 0) [f(x+h) - f(x)] / h for f(x) = $25x^2$.

Rule/Reference: `f'(x) = 2({idx})x`

Derivative at a Point

EASY 1. Evaluate f'(0) for f(x) = $1x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 2. Evaluate f'(1) for f(x) = $2x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 3. Evaluate f'(2) for f(x) = $3x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 4. Evaluate f'(3) for f(x) = $4x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 5. Evaluate f'(4) for f(x) = $5x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 6. Evaluate f'(0) for f(x) = $6x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 7. Evaluate f'(1) for f(x) = $7x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

EASY 8. Evaluate f'(2) for f(x) = $8x^2$ + 5x.

Rule/Reference: `f'(x) = 2({idx})x + 5`

INTERM. 9. Evaluate f'(3) for $f(x) = 9x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 10. Evaluate f'(4) for $f(x) = 10x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 11. Evaluate f'(0) for $f(x) = 11x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 12. Evaluate f'(1) for $f(x) = 12x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 13. Evaluate f'(2) for $f(x) = 13x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 14. Evaluate f'(3) for $f(x) = 14x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 15. Evaluate f'(4) for $f(x) = 15x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 16. Evaluate f'(0) for $f(x) = 16x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

INTERM. 17. Evaluate f'(1) for $f(x) = 17x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 18. Evaluate f'(2) for $f(x) = 18x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 19. Evaluate f'(3) for $f(x) = 19x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 20. Evaluate f'(4) for $f(x) = 20x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 21. Evaluate f'(0) for $f(x) = 21x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 22. Evaluate f'(1) for $f(x) = 22x^2 + 5x$.

Rule/Reference: f'(x) = 2({idx})x + 5

DIFFICULT 23. Evaluate f'(2) for $f(x) = 23x^2 + 5x$.

Rule/Reference: `f'(x) = 2({idx})x + 5`

DIFFICULT 24. Evaluate f'(3) for $f(x) = 24x^2 + 5x$.

Rule/Reference: `f'(x) = 2({idx})x + 5`

DIFFICULT 25. Evaluate f'(4) for $f(x) = 25x^2 + 5x$.

Rule/Reference: `f'(x) = 2({idx})x + 5`

Power Rules

EASY 1. Find f'(x) for $f(x) = 1x^2$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 2. Find f'(x) for $f(x) = 2x^3$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 3. Find f'(x) for $f(x) = 3x^4$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 4. Find f'(x) for $f(x) = 4x^5$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 5. Find f'(x) for $f(x) = 5x^6$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 6. Find f'(x) for $f(x) = 6x^2$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 7. Find f'(x) for $f(x) = 7x^3$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

EASY 8. Find f'(x) for $f(x) = 8x^4$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

INTERM. 9. Find f'(x) for $f(x) = 9x^5$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

INTERM. 10. Find f'(x) for $f(x) = 10x^6$.

Rule/Reference: $d/dx\ (ax^n) = n * a * x^{n-1}$

INTERM. 11. Find f'(x) for $f(x) = 11x^2$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 12. Find f'(x) for $f(x) = 12x^3$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 13. Find f'(x) for $f(x) = 13x^4$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 14. Find f'(x) for $f(x) = 14x^5$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 15. Find f'(x) for $f(x) = 15x^6$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 16. Find f'(x) for $f(x) = 16x^2$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

INTERM. 17. Find f'(x) for $f(x) = 17x^3$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 18. Find f'(x) for $f(x) = 18x^4$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 19. Find f'(x) for $f(x) = 19x^5$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 20. Find f'(x) for $f(x) = 20x^6$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 21. Find f'(x) for $f(x) = 21x^2$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 22. Find f'(x) for $f(x) = 22x^3$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 23. Find f'(x) for $f(x) = 23x^4$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 24. Find f'(x) for $f(x) = 24x^5$.

Rule/Reference: d/dx (ax^n) = n * a * x^{n-1}

DIFFICULT 25. Find f'(x) for $f(x) = 25x^6$.

Rule/Reference: `d/dx (ax^n) = n * a * x^(n-1)`

Derivative Rules: Constant, Sum, Difference, Scalar

EASY 1. Differentiate $f(x) = 1x^3 + 2x - 0$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 2. Differentiate $f(x) = 2x^3 + 3x - 1$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 3. Differentiate $f(x) = 3x^3 + 4x - 2$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 4. Differentiate $f(x) = 4x^3 + 5x - 3$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 5. Differentiate $f(x) = 5x^3 + 6x - 4$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 6. Differentiate $f(x) = 6x^3 + 7x - 5$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 7. Differentiate $f(x) = 7x^3 + 8x - 6$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

EASY 8. Differentiate $f(x) = 8x^3 + 9x - 7$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

INTERM. 9. Differentiate $f(x) = 9x^3 + 10x - 8$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

INTERM. 10. Differentiate $f(x) = 10x^3 + 11x - 9$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

INTERM. 11. Differentiate $f(x) = 11x^3 + 12x - 10$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

INTERM. 12. Differentiate $f(x) = 12x^3 + 13x - 11$.

Rule/Reference: `d/dx [g(x) + h(x)] = g'(x) + h'(x)`

INTERM. 13. Differentiate $f(x) = 13x^3 + 14x - 12$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

INTERM. 14. Differentiate $f(x) = 14x^3 + 15x - 13$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

INTERM. 15. Differentiate $f(x) = 15x^3 + 16x - 14$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

INTERM. 16. Differentiate $f(x) = 16x^3 + 17x - 15$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

INTERM. 17. Differentiate $f(x) = 17x^3 + 18x - 16$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 18. Differentiate $f(x) = 18x^3 + 19x - 17$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 19. Differentiate $f(x) = 19x^3 + 20x - 18$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 20. Differentiate $f(x) = 20x^3 + 21x - 19$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 21. Differentiate $f(x) = 21x^3 + 22x - 20$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 22. Differentiate $f(x) = 22x^3 + 23x - 21$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 23. Differentiate $f(x) = 23x^3 + 24x - 22$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 24. Differentiate $f(x) = 24x^3 + 25x - 23$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

DIFFICULT 25. Differentiate $f(x) = 25x^3 + 26x - 24$.

Rule/Reference: d/dx [g(x) + h(x)] = g'(x) + h'(x)

Equation of Tangent Line

EASY 1. Find the equation of the tangent line to $f(x) = x^2$ at $x = 0$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 2. Find the equation of the tangent line to $f(x) = x^2$ at $x = 1$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 3. Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 4. Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 5. Find the equation of the tangent line to $f(x) = x^2$ at $x = 4$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 6. Find the equation of the tangent line to $f(x) = x^2$ at $x = 0$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 7. Find the equation of the tangent line to $f(x) = x^2$ at $x = 1$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

EASY 8. Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 9. Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 10. Find the equation of the tangent line to $f(x) = x^2$ at $x = 4$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 11. Find the equation of the tangent line to $f(x) = x^2$ at $x = 0$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 12. Find the equation of the tangent line to $f(x) = x^2$ at $x = 1$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 13. Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 14. Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 15. Find the equation of the tangent line to $f(x) = x^2$ at $x = 4$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 16. Find the equation of the tangent line to $f(x) = x^2$ at $x = 0$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

INTERM. 17. Find the equation of the tangent line to $f(x) = x^2$ at $x = 1$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 18. Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 19. Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 20. Find the equation of the tangent line to $f(x) = x^2$ at $x = 4$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 21. Find the equation of the tangent line to $f(x) = x^2$ at $x = 0$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 22. Find the equation of the tangent line to $f(x) = x^2$ at $x = 1$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 23. Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 24. Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

DIFFICULT 25. Find the equation of the tangent line to $f(x) = x^2$ at $x = 4$.

Rule/Reference: y - f(a) = f'(a) * (x - a)

Trigonometric, Logarithmic, and Exponential Functions

EASY 1. Find f'(x) for f(x) = 1sin(x) + 1ln(x).

Rule/Reference: d/dx sin(x) = cos(x)

EASY 2. Find f'(x) for $f(x) = 2e^x - 2\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

EASY 3. Find f'(x) for $f(x) = 3\log10(x) + 3x^2$.

Rule/Reference: d/dx ln(x) = 1/x

EASY 4. Find f'(x) for $f(x) = 4\sin(x) + 4\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

EASY 5. Find f'(x) for $f(x) = 5e^x - 5\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

EASY 6. Find f'(x) for $f(x) = 6\log10(x) + 6x^2$.

Rule/Reference: d/dx ln(x) = 1/x

EASY 7. Find f'(x) for $f(x) = 7\sin(x) + 7\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

EASY 8. Find f'(x) for $f(x) = 8e^x - 8\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

INTERM. 9. Find f'(x) for $f(x) = 9\log10(x) + 9x^2$.

Rule/Reference: d/dx ln(x) = 1/x

INTERM. 10. Find f'(x) for $f(x) = 10\sin(x) + 10\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

INTERM. 11. Find f'(x) for $f(x) = 11e^x - 11\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

INTERM. 12. Find f'(x) for $f(x) = 12\log10(x) + 12x^2$.

Rule/Reference: d/dx ln(x) = 1/x

INTERM. 13. Find f'(x) for $f(x) = 13\sin(x) + 13\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

INTERM. 14. Find f'(x) for $f(x) = 14e^x - 14\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

INTERM. 15. Find f'(x) for $f(x) = 15\log10(x) + 15x^2$.

Rule/Reference: d/dx ln(x) = 1/x

INTERM. 16. Find f'(x) for $f(x) = 16\sin(x) + 16\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

INTERM. 17. Find f'(x) for $f(x) = 17e^x - 17\cos(x)$.

Rule/Reference: d/dx $e^x = e^x$

DIFFICULT 18. Find f'(x) for $f(x) = 18\log10(x) + 18x^2$.

Rule/Reference: d/dx ln(x) = 1/x

DIFFICULT 19. Find f'(x) for $f(x) = 19\sin(x) + 19\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

DIFFICULT 20. Find f'(x) for $f(x) = 20e^{x} - 20\cos(x)$.

Rule/Reference: d/dx $e^{x} = e^{x}$

DIFFICULT 21. Find f'(x) for $f(x) = 21\log10(x) + 21x^2$.

Rule/Reference: d/dx ln(x) = 1/x

DIFFICULT 22. Find f'(x) for $f(x) = 22\sin(x) + 22\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

DIFFICULT 23. Find f'(x) for $f(x) = 23e^{x} - 23\cos(x)$.

Rule/Reference: d/dx $e^{x} = e^{x}$

DIFFICULT 24. Find f'(x) for $f(x) = 24\log10(x) + 24x^2$.

Rule/Reference: d/dx ln(x) = 1/x

DIFFICULT 25. Find f'(x) for $f(x) = 25\sin(x) + 25\ln(x)$.

Rule/Reference: d/dx sin(x) = cos(x)

Product Rules

EASY 1. Find f'(x) for $f(x) = (1x^2)(2x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 2. Find f'(x) for $f(x) = (2x^2)(3x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 3. Find f'(x) for $f(x) = (3x^2)(4x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 4. Find f'(x) for $f(x) = (4x^2)(5x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 5. Find f'(x) for $f(x) = (5x^2)(6x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 6. Find f'(x) for $f(x) = (6x^2)(7x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 7. Find f'(x) for $f(x) = (7x^2)(8x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

EASY 8. Find f'(x) for $f(x) = (8x^2)(9x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 9. Find f'(x) for $f(x) = (9x^2)(10x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 10. Find f'(x) for $f(x) = (10x^2)(11x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 11. Find f'(x) for $f(x) = (11x^2)(12x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 12. Find f'(x) for $f(x) = (12x^2)(13x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 13. Find f'(x) for $f(x) = (13x^2)(14x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 14. Find f'(x) for $f(x) = (14x^2)(15x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 15. Find f'(x) for $f(x) = (15x^2)(16x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 16. Find f'(x) for $f(x) = (16x^2)(17x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

INTERM. 17. Find f'(x) for $f(x) = (17x^2)(18x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 18. Find f'(x) for $f(x) = (18x^2)(19x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 19. Find f'(x) for $f(x) = (19x^2)(20x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 20. Find f'(x) for $f(x) = (20x^2)(21x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 21. Find f'(x) for $f(x) = (21x^2)(22x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 22. Find f'(x) for $f(x) = (22x^2)(23x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 23. Find f'(x) for $f(x) = (23x^2)(24x^3)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 24. Find f'(x) for $f(x) = (24x^2)(25x^4)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

DIFFICULT 25. Find f'(x) for $f(x) = (25x^2)(26x^2)$.

Rule/Reference: d/dx [u(x)v(x)] = u'(x)v(x) + u(x)v'(x)

Chain Rule

EASY 1. Find f'(x) for $f(x) = (1x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 2. Find f'(x) for $f(x) = (2x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 3. Find f'(x) for $f(x) = (3x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 4. Find f'(x) for $f(x) = (4x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 5. Find f'(x) for $f(x) = (5x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 6. Find f'(x) for $f(x) = (6x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 7. Find f'(x) for $f(x) = (7x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

EASY 8. Find f'(x) for $f(x) = (8x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 9. Find f'(x) for $f(x) = (9x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 10. Find f'(x) for $f(x) = (10x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 11. Find f'(x) for $f(x) = (11x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 12. Find f'(x) for $f(x) = (12x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 13. Find f'(x) for $f(x) = (13x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 14. Find f'(x) for $f(x) = (14x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 15. Find f'(x) for $f(x) = (15x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 16. Find f'(x) for $f(x) = (16x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

INTERM. 17. Find f'(x) for $f(x) = (17x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 18. Find f'(x) for $f(x) = (18x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 19. Find f'(x) for $f(x) = (19x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 20. Find f'(x) for $f(x) = (20x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 21. Find f'(x) for $f(x) = (21x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 22. Find f'(x) for $f(x) = (22x + 1)^3$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 23. Find f'(x) for $f(x) = (23x + 1)^4$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 24. Find f'(x) for $f(x) = (24x + 1)^5$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)

DIFFICULT 25. Find f'(x) for $f(x) = (25x + 1)^2$.

Rule/Reference: d/dx [g(h(x))] = g'(h(x)) * h'(x)